

ONTOLOGY-COMPARISON

Ontology Comparison: Record Harm vs. Related Frameworks

This document maps the Record Harm Ontology to established frameworks in information security, archival science, and diplomatics.

Overview

The Record Harm Ontology focuses on **ontological attacks on records** — how records can be damaged, destroyed, or corrupted as informational entities. While it shares concerns with other frameworks, its scope and granularity differ:

- **STRIDE** (threat modeling): Focuses on system security threats
- **CIA Triad** (information security): Focuses on confidentiality, integrity, availability
- **InterPARES Diplomatics**: Focuses on record authenticity and trustworthiness
- **Spoliation Doctrine** (legal): Focuses on evidence destruction and concealment

Comparison Table

Record Harm ↔ STRIDE

Record Harm	STRIDE Category	Notes
Fabrication	Tampering	Both involve creating/modifying false information
Alteration	Tampering	Direct modification of existing data
Destruction	Denial of Service (partial)	DoS is transient unavailability; Destruction is permanent loss
Suppression	Denial of Service (partial)	Suppression is deliberate concealment; DoS is service disruption
Denial	Repudiation	Both involve denying validity/authorship
Omission	(no direct mapping)	STRIDE has no completeness category. Not an inverse of Information Disclosure: omission attacks what the record contains, disclosure attacks who can read it — different aspects, not opposite ends of one.
ForgeryOfProvenance	Spoofing	Both involve falsifying identity/origin
Repudiation	Repudiation	Direct mapping

Record Harm	STRIDE Category	Notes
Contamination	Tampering	Mixing genuine with false
Decontextualization	<i>(no direct mapping)</i>	STRIDE doesn't model context stripping
Fragmentation	<i>(no direct mapping)</i>	STRIDE doesn't model record fragmentation
Obfuscation	<i>(no direct mapping)</i>	STRIDE doesn't model deliberate obscurity
<i>(no mapping)</i>	Information Disclosure	Deliberate gap: Leaked records are undamaged as records
<i>(no mapping)</i>	Elevation of Privilege	Outside record harm scope (system access, not record integrity)

Key Differences:

- **STRIDE is system-centric**; Record Harm is **record-centric**
- **STRIDE includes confidentiality** (Information Disclosure); Record Harm **deliberately excludes it** — a leaked record is authentic, complete, and accessible
- **Record Harm distinguishes Suppression from Destruction**; STRIDE conflates both under Denial of Service
- **Record Harm models composite harms**; STRIDE categories are flat

Record Harm ↔ CIA Triad

Record Harm	CIA Category	Notes
Destruction	Availability	Permanent loss vs. transient unavailability
Suppression	Availability	Deliberate concealment vs. system failure
Fabrication	Integrity	Creating false records
Alteration	Integrity	Modifying existing records
Contamination	Integrity	Mixing genuine with false
Omission	Integrity	Incomplete records
ForgeryOfProvenance	Integrity + Authenticity	Falsifying origin/custody
Denial	<i>(no mapping)</i>	Contesting legitimacy. Trustworthiness is a Record Harm aspect, not a CIA category — see the key difference below.
Repudiation	<i>(no mapping)</i>	Disavowing authorship. Same gap as Denial.
Decontextualization	Integrity	Stripping metadata/context

Record Harm	CIA Category	Notes
Fragmentation	Integrity + Availability	Breaking into disconnected pieces
Obfuscation	Availability	Degrading comprehensibility
<i>(no mapping)</i>	Confidentiality	Deliberate gap: See STRIDE notes above

Key Differences:

- **CIA is property-centric** (what properties are violated); Record Harm is **action-centric** (what attacks occur)
- **CIA Integrity is broad**; Record Harm **separates what it lumps together** — **Alteration** and **Omission** both violate integrity but are distinct primes, and **Fabrication** is split off entirely as an attack on Authenticity
- **CIA Availability conflates** destruction, suppression, and obfuscation; Record Harm **separates** them
- **Record Harm adds Trustworthiness** as a distinct aspect beyond CIA

Record Harm ↔ InterPARES Diplomatics

InterPARES (International Research on Permanent Authentic Records in Electronic Systems) provides a framework for archival authenticity.

Record Harm	Diplomatics Concept	Notes
Authenticity (aspect)	Authenticity	Core shared concept
Context (aspect)	Archival Bond	Both emphasize provenance and relationships
Integrity (aspect)	Completeness	Both require wholeness
Fabrication	Forgery	Creating false records
Alteration	Corruption	Unauthorized modification
ForgeryOfProvenance	False Provenance	Falsifying archival context
Decontextualization	Loss of Archival Bond	Severing from context
Omission	Incompleteness	Missing elements
Destruction	Loss	Permanent removal
Suppression	Concealment	Deliberate hiding
Denial	Contestation of Authenticity	Challenging legitimacy
Fragmentation	Dispersal	Breaking archival unity
Contamination	Adulteration	Mixing genuine with false

Record Harm	Diplomatics Concept	Notes
Obfuscation	Obscuration	Degrading intelligibility
Repudiation	Disavowal of Authorship	Denying creation

Key Similarities:

- Both emphasize **authenticity and provenance**
- Both recognize **context as essential** to record meaning
- Both distinguish **form from content**

Key Differences:

- **Diplomatics is preservation-focused**; Record Harm is **attack-focused**
- **Diplomatics emphasizes archival bond** (relationships between records); Record Harm **models this via hasElement**
- **Diplomatics is community-calibrated**; Record Harm **includes inside-agent records** (memories, beliefs)

Record Harm ↔ Spoliation Doctrine (Legal)

Spoliation doctrine addresses destruction or concealment of evidence in legal proceedings.

Record Harm	Spoliation Concept	Notes
Destruction	Destruction of Evidence	Direct mapping
Suppression	Concealment/Withholding	Hiding evidence from discovery
Alteration	Tampering with Evidence	Modifying evidence
Fabrication	Fabrication of Evidence	Creating false evidence
Omission	Selective Production	Withholding relevant documents
ForgeryOfProvenance	False Chain of Custody	Falsifying evidence handling
Contamination	Tainting Evidence	Mixing genuine with false
Denial	Contesting Authenticity	Challenging evidence validity
Repudiation	Disavowal	Denying authorship/authority
Decontextualization	Selective Quotation	Removing context to mislead
Fragmentation	Partial Production	Producing incomplete sets
Obfuscation	Burying in Volume	Making evidence hard to find

Key Similarities:

- Both recognize **destruction and concealment as distinct wrongs**
- Both address **chain of custody** (ForgeryOfProvenance)
- Both recognize **selective production** (Omission)

Key Differences:

- **Spoliation is adversarial** (litigation context); Record Harm is **general-purpose**
- **Spoliation focuses on intent and bad faith**; Record Harm **models the acts themselves**
- **Spoliation has legal remedies** (adverse inference, sanctions); Record Harm is **descriptive, not prescriptive**

Unique Contributions of Record Harm Ontology

1. Inside-Agent Records (v3.1)

No other framework explicitly models:

- Memories as records
- Beliefs as records
- Self-directed harm (repression, self-deception, motivated forgetting)
- The agent/community boundary

Example: A witness repressing a memory is `ex:Suppression` where `perpetrator = bearer`. This is inexpressible in STRIDE, CIA, diplomatics, or spoliation.

2. Formal Dependency Structure

Record Harm uniquely provides:

- Distinction between **prime** (irreducible) and **composite** (derived) harms
- Explicit `buildsUpon` relationships
- Reasoner-derivable composite membership
- Transitive closure via SPARQL property paths

Example: `ex:Obfuscation buildsUpon ex:Suppression, ex:Alteration` — neither STRIDE nor CIA models this dependency.

3. Mereological Structure (v3.1)

Record Harm uniquely models:

- Records as composed of other records (`hasElement`)
- Community registers as records in their own right
- Non-admission as `Omission` from the register

Example: A parish register whose elements are individual testimonies. Refusing to admit a testimony is `ex:Omission` from that register.

4. Aspect-Based Targeting

Record Harm provides:

- 6 fundamental aspects (Existence, Authenticity, Integrity, Accessibility, Context, Trustworthiness)
- Multi-valued targeting (one harm can attack multiple aspects)
- Coverage guarantee (every aspect reachable from at least one prime)

Example: `ex:Alteration` targets both `Authenticity` and `Integrity` — CIA would only say "Integrity violation."

5. Event Layer with Severity

Record Harm separates:

- Harm **types** (abstract categories)
- Harm **events** (specific occurrences with dates, perpetrators, severity)

Example: Two `ex:Destruction` events can have different severities (destroying a parking receipt vs. a birth certificate).

Integration Guidance

For Information Security Practitioners

- **Map STRIDE threats to Record Harms** for evidence integrity analysis
- **Use Record Harm for forensics** where CIA Triad is too coarse
- **Note the confidentiality gap** — add your own confidentiality layer if needed

For Archivists and Records Managers

- **Use Record Harm alongside diplomatics** for threat modeling
- **Model archival bonds via** `hasElement` for collection integrity
- **Track provenance attacks via** `ForgeryOfProvenance`

For Legal Professionals

- **Map spoliation claims to Record Harms** for structured analysis
- **Use severity scoring** for proportionality arguments
- **Track chain of custody via** `ForgeryOfProvenance` events

For AI Ethics and Misinformation Research

- **Model disinformation campaigns as** `HarmPattern` instances
- **Track inside-agent harms** (belief manipulation, memory distortion)

- Use `Contamination` for genuine/false mixing in information ecosystems

Limitations and Scope Boundaries

What Record Harm Does NOT Cover

1. **Confidentiality breaches** — Deliberate exclusion (see notes above)
2. **System vulnerabilities** — STRIDE is better for this
3. **Access control** — Outside record harm scope
4. **Encryption/cryptography** — Technical implementation, not ontological harm
5. **Pure non-creation** — Witnessing without recording (see `ex:RecordHarm` scope note)

What Record Harm DOES Cover

1. **Ontological attacks on records** — Existence, authenticity, integrity, accessibility, context, trustworthiness
2. **Inside-agent and inside-community records** — Memories, beliefs, registers, archives
3. **Self-directed harm** — Repression, self-deception, motivated forgetting
4. **Composite harm structures** — Dependencies and patterns
5. **Event tracking** — Specific occurrences with severity and attribution

References

- **STRIDE**: Microsoft Threat Modeling Framework
- **CIA Triad**: Confidentiality, Integrity, Availability (foundational InfoSec model)
- **InterPARES**: International Research on Permanent Authentic Records in Electronic Systems
- **Spoliation Doctrine**: Legal doctrine on evidence destruction/concealment
- **Record Harm Ontology**: This ontology (v3.1)

Versioning

This document describes the Record Harm Ontology at **v3.1** and carries no version of its own. The ontology has a single version, asserted in `owl:versionInfo` and recorded in [CHANGELOG.md](#); a separate per-document number would be a second version axis that drifts out of step with the thing it describes — the same failure mode as v1's `isFundamentalRoot` boolean drifting out of step with class membership.

When the mappings below change because the taxonomy changed, record it in the ontology's changelog entry for that release, not here.